

CONTADA / SMART CENTRE TECHNICAL SUMMARY

Project Context: Embedded systems engineering co-op with the Conestoga SMART Research Centre in collaboration with Purus Power. The work supported the CONTADA wind-energy and battery-storage research platform through edge-device telemetry, sensor integration, and control-system validation workflows.

ROLE AND CONTRIBUTION

- Designed and tested edge-device network concepts using ESP32-S3, STM32, Raspberry Pi 5, and networked telemetry for a renewable-energy research platform.
- Developed and modified embedded firmware for microcontroller-based sensor acquisition, structured data publishing, and system-status monitoring.
- Integrated environmental and motion sensors, including BME680, BNO085, and ADXL335 devices, into firmware and telemetry workflows.
- Connected embedded devices to Linux-hosted dashboards using MQTT, Raspberry Pi infrastructure, and Node-RED visualization for real-time system feedback.
- Created MATLAB/Simulink models to simulate and communicate sensor-driven control behavior during embedded system development.
- Presented weekly engineering updates to Purus Power and SMART Centre teams, translating technical progress, risks, and next steps into clear action items.

TECHNICAL SCOPE

Area	Relevant Work
Embedded Platforms	ESP32-S3 and STM32 edge devices for sensor acquisition, firmware testing, and telemetry prototyping.
Sensors & Interfaces	BME680, BNO085, ADXL335, I2C-style sensor workflows, analog sensor acquisition, and embedded peripheral integration.
Telemetry & Backend	MQTT data publishing, Raspberry Pi 5 backend infrastructure, Node-RED dashboards, JSON-style telemetry, and Linux-based troubleshooting.
Modeling & Validation	MATLAB/Simulink models used to simulate, validate, and communicate sensor-driven control behavior.
Testing & Debugging	Hardware/software integration testing, serial debugging, sensor verification, telemetry checks, and iterative root-cause troubleshooting.

RELEVANCE TO EMBEDDED SOFTWARE ROLES

- Strengthened practical experience debugging across firmware, hardware, communication links, and Linux-based visualization tools.
- Built experience communicating embedded-system progress to both technical and non-technical stakeholders in a collaborative engineering environment.
- Developed a system-level mindset by connecting microcontroller firmware, sensors, backend telemetry, modeling tools, and validation documentation.

Confidentiality Note: This summary is intentionally limited to non-confidential, high-level technical responsibilities and does not disclose proprietary design details, client-specific intellectual property, or sensitive research data.